

DASHBOARD DOCUMENTATION

Global Superstore Sales Analysis

Developed in Microsoft Power BI Desktop
2026

1. Executive Summary

This document provides a comprehensive technical and analytical record of the Global Superstore Sales Analysis dashboard, built using Microsoft Power BI Desktop. The dashboard transforms raw transactional retail data into actionable intelligence across dimensions of product category, customer segment, geography, and time, enabling data-driven decision-making for commercial strategy.

The dashboard covers a four-year operational period (2011–2014), spanning global retail activity across multiple regions, and delivers a single-page interactive reporting interface with real-time filter capabilities.

Total Profit	Total Sales	Profit Margin	Sales Last Year	Total Orders
\$545.2K	\$3.9M	13.87%	\$3.9M	8,031

2. Project Overview

2.1 Objectives

- Consolidate global sales performance data into a unified, interactive single-page dashboard
- Enable year-on-year and period-on-period comparison of sales, profit and margin
- Identify top-performing and underperforming product sub-categories
- Surface regional and country-level disparities in profitability and order volume
- Support strategic decisions on product mix, market focus, and operational priorities

2.2 Data Source

The analysis is based on the Global Superstore dataset, a widely used benchmark retail dataset containing transactional records across product categories, customer segments, countries, and shipping details.

Attribute	Details
Dataset	Global Superstore (Retail Transactional Data)

Tool	Microsoft Power BI Desktop
Time Period	2011 – 2014 (4 years)
Scope	Global — multiple regions, countries, segments
Granularity	Order-level transactions
Key Dimensions	Category, Sub-Category, Segment, Region, Country, Year, Month, Order Priority

2.3 Tools & Technologies

- Microsoft Power BI Desktop — primary development environment
- DAX (Data Analysis Expressions) — calculated measures and KPI logic
- Power Query (M Language) — data transformation and cleansing
- Custom visualisations — bar charts, line charts, donut charts, geographic maps, Pareto tables

3. Dashboard Structure & Components

The dashboard is designed as a single-page report layout, maximising information density while maintaining visual clarity. All components respond dynamically to the Year/Month filter slicer in the top-left panel.

3.1 Global KPI Header

A five-metric KPI strip anchors the top of the dashboard, providing an at-a-glance summary of business performance:

- Total Profit — \$545.2K: Net profit generated across all transactions
- Total Sales — \$3.9M: Gross revenue across all categories and regions
- Profit Margin — 13.87%: Overall profitability ratio
- Sales Last Year — \$3.9M: Year-over-year sales benchmark
- Total Orders — 8,031: Volume of individual orders processed

3.2 Profit Margin by Year (Line Chart)

A line chart tracking annual profit margin from 2011 to 2014. Key insight: margin dipped to approximately 13.1% in 2012 before recovering to approximately 14.5% by 2014, indicating improving cost management or pricing optimisation over the period.

3.3 Category Performance Table

A tabular breakdown of the three product categories by orders, sales, and profit:

Category	Total Orders	Total Sales	Total Profit	Profit Margin
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Furniture	1,891	\$1,343,549.6	\$117,375.0	8.74%
Office Supplies	4,102	\$1,040,744.4	\$185,769.5	17.85%
Technology	2,038	\$1,547,533.3	\$242,060.6	15.64%
Total	8,031	\$3,931,827.2	\$545,205.0	13.87%

Technology leads on both total sales and profit despite having fewer orders than Office Supplies. Furniture, despite high sales volume, delivers the lowest profit margin, warranting strategic review.

3.4 Profit Margin by Category (Donut Chart)

A donut chart illustrates relative profit contribution by category:

- Technology: 17.85% margin — highest among all categories
- Office Supplies: 15.64% margin — strong and stable
- Furniture: 8.74% margin — significantly below portfolio average; loss-making at sub-category level for Tables (–\$9K)

3.5 Total Profit by Sub-Category (Bar Chart)

A ranked bar chart reveals profit distribution across 17 sub-categories. Top performers and areas of concern:

Sub-Category	Total Profit	Notes
Phones	\$85K	Highest profit sub-category
Copiers	\$85K	Joint highest
Appliances	\$72K	Strong performer
Bookcases	\$67K	Top furniture item
Chairs	\$44K	Mid-tier
Accessories	\$41K	Mid-tier
Binders	\$31K	Mid-tier
Storage	\$31K	Mid-tier
Machines	\$30K	Mid-tier
Tables	(\$9K)	Only loss-making sub-category

3.6 Profit Margin by Segment (Donut Chart)

Customer segment profitability analysis:

- Consumer: 14.90% margin — highest segment margin
- Corporate: 14.65% margin — near-average performance
- Home Office: 13.53% margin — lowest, suggesting potential pricing or mix issues

3.7 Total Profit by Order Priority (Horizontal Bar)

Profit attributed by order fulfilment priority level:

- Medium priority: \$245.4K — accounts for the largest share
- High priority: \$177.1K
- Critical priority: \$68.2K
- Low priority: \$22.0K

The dominance of Medium and High priority orders indicates that the business's primary revenue base is not dependent on expedited fulfilment, providing operational scheduling flexibility.

3.8 Average Shipping Cost by Country

A bar chart identifies countries with the highest average shipping costs. Notable findings:

- Tanzania — highest average shipping cost at \$763
- Taiwan — \$635
- Saudi Arabia — \$493
- China — \$390

High shipping costs in emerging markets may be depressing net profitability. This visualisation supports supply chain optimisation decisions.

3.9 Regional Performance Summary

The dashboard's geographic panel presents a segmented breakdown by global trading region:

Region	Total Orders	Total Sales	Total Profit	Profit Margin
Africa	705	\$313,123.5	\$59,055.4	18.86%
Canada	65	\$26,167.2	\$7,840.2	29.96%
Caribbean	246	\$88,318.9	\$12,295.7	13.92%
Central	1,608	\$810,705.0	\$104,846.5	12.93%
Central Asia	331	\$209,611.4	\$33,628.8	16.04%
East	421	\$192,660.0	\$33,938.8	17.62%
EMEA	753	\$298,225.1	\$40,355.2	13.53%

Canada stands out with the highest profit margin at 29.96%, though on a small order base of 65 orders. Central America has the highest absolute profit (\$104.8K) driven by its large order volume of 1,608. EMEA and Caribbean deliver near-average margins.

3.10 Total Sales by Country (Bar Chart)

A ranked bar chart of the top 10 countries by total sales:

Country	Total Sales	Rank
United States	\$0.66M	1st
Australia	\$0.31M	2nd
France	\$0.26M	3rd
Mexico	\$0.23M	4th
China	\$0.20M	5th
Germany	\$0.17M	6th
United Kingdom	\$0.16M	7th
India	\$0.15M	8th
Indonesia	\$0.12M	9th
Brazil	\$0.11M	10th

The United States dominates with \$0.66M in sales, more than twice the second-ranked market (Australia at \$0.31M), highlighting geographic concentration risk.

3.11 Product Pareto Analysis Table

A Pareto table quantifies cumulative sales contribution by category:

- Technology: \$1,547,533 — top revenue generator (Pareto % = 0.39)
- Furniture: \$1,343,549 — second (cumulative = 0.74)
- Office Supplies: \$1,040,744 — third (cumulative = 1.00)

Office Supplies delivers the highest number of orders (4,102) but lowest total sales, suggesting a high-volume, low-value transaction profile consistent with consumable products.

3.12 Interactive Year/Month Filter Slicer

A hierarchical filter panel on the left allows users to drill down by year (2011–2015) and month. This slicer propagates across all dashboard visuals simultaneously, enabling period-specific analysis without switching report pages.

4. Key Analytical Insights

4.1 Profitability Trends

- Overall profit margin of 13.87% is healthy for retail, with a positive trend from 2012 to 2014
- Technology is the highest-margin category at 17.85%, making it the most strategically valuable product line
- Furniture's Tables sub-category is the sole loss-making segment at –\$9K, representing a pricing or cost structure problem

4.2 Geographic Opportunities

- Canada shows a 29.96% profit margin — highest globally — suggesting strong pricing power in a low-competition market
- The Central region contributes the most absolute profit (\$104.8K) due to scale
- High shipping costs in Tanzania and other emerging markets may be suppressing addressable profit in growth markets

4.3 Product Mix Recommendations

- Phones and Copiers each deliver \$85K profit — investing in expanding these product lines would have the highest profit impact
- Tables should be reviewed for pricing adjustment, supplier renegotiation, or potential discontinuation
- Office Supplies drives volume (4,102 orders) but comparatively lower unit profitability; margin improvement here could significantly impact bottom line

4.4 Segment Targeting

- Consumer segment yields the best margin (14.90%), suggesting that B2C channels may offer more favourable pricing dynamics
- Home Office segment underperforms at 13.53% — a targeted pricing or bundling strategy could improve returns

5. Technical Design & Implementation

5.1 DAX Measures

Core calculated measures developed for the dashboard:

Measure	DAX Formula (Conceptual)	Purpose
Total Profit	SUM(Product Table[Profit])	Aggregate net profit
Total Sales	SUM(Product Table [Sales])	Aggregate gross revenue
Profit Margin	DIVIDE([Total Profit], [Total Sales])	Ratio calculation
Sales Last Year	CALCULATE([Total Sales], SAMEPERIODLASTYEAR(...))	YoY benchmark
Total Orders	COUNTROWS(Product Table)	Order volume count

5.2 Data Model

- Star schema design with Orders as the central fact table
- Dimension tables for: Product (Category, Sub-Category), Geography (Region, Country), Date (Year, Month), Customer (Segment)
- Relationships defined on Order ID, Product ID, Customer ID, and Date keys

- Date table created using CALENDAR() function to enable time intelligence

5.3 Visualisation Design Choices

- Consistent purple palette used throughout for professional brand coherence
- Donut charts used for part-to-whole relationships (category and segment margins)
- Ranked bar charts preferred over pie charts for sub-category comparisons to support precise magnitude reading
- Geographic heat map used to contextualise country-level sales data spatially
- KPI cards in header zone following dashboard design best practice (most important metrics first)

5.4 Performance Optimisation

- Measures use DIVIDE() instead of division operator to handle zero denominators gracefully
- Filter context managed via CALCULATE() for period comparisons
- Visual interactions configured to cross-filter (not cross-highlight) for analytical clarity

6. Skills Demonstrated

Skill Area	Demonstrated Competencies
Data Visualisation	Designed and built a multi-component Power BI dashboard with 10+ visualisations
Business Intelligence	Translated raw transactional data into strategic KPIs and trend insights
DAX Programming	Developed calculated measures including time intelligence and ratio metrics
Data Modelling	Designed a star schema relational data model optimised for reporting
Analytical Thinking	Identified actionable insights across product, segment, and geography dimensions
Dashboard Design	Applied UX principles: visual hierarchy, colour consistency, filter interactivity
Reporting	Produced clear, stakeholder-ready documentation and structured findings

7. Conclusions & Recommendations

The Global Superstore Sales Dashboard successfully delivers a comprehensive, interactive view of a global retail operation's performance across product, customer, and geographic dimensions. Key strategic recommendations arising from the analysis:

- Prioritise Technology product investment, particularly in Phones and Copiers, as the highest-return sub-categories
- Conduct a strategic review of the Furniture category — specifically Tables — to address the only loss-making sub-category in the portfolio
- Expand operations in Canada, leveraging the 29.96% profit margin as a market validation signal
- Investigate and optimise shipping logistics in Tanzania and other high-shipping-cost markets to unlock latent profitability
- Target Consumer segment customers with premium product offerings, given their demonstrated higher profit margin
- Implement quarterly dashboard refresh cycles to monitor trend continuation into 2015 and beyond

This project demonstrates the capacity to design, build, and interpret a production-grade business intelligence solution, translating complex datasets into clear, decision-ready insights.